# Shannon's Information Theory & Decision Tree

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## Information Theory

Claude Shannon (1916– 2001) introduced a mathematical framework for **quantifying information** - **Information Theory**.

This poster covers the basics of Information Theory – **Information and Entropy** – as well as exploring the classification algorithm derived from Shannon's Information Theory – **Decision Tree.**

## Information $I(X)$

How can we measure the amount of "information" in a message or event? Shannon states that:

- Rare or unexpected events give more information.
- Common or expected events give less information.

$$\textbf{Information} = I(X) = -log_2(p_i)$$

If an event is unlikely ($p$ is small), it holds more information.
If an event is very likely ($p$ is large), it holds less information.

## Game of 64 - Binary Search

- I choose a random number between 1 and 64
- I can only answer yes/no questions
- How many questions do you need to find out what it is?
- How much information do you get if you pick random numbers?

## Game of 64 - Binary Search

- **1 is wrong:** New information is $-\log_2\left(\frac{63}{64}\right) \approx 0.0227$ bits.
- **2 is wrong:** New information is $-\log_2\left(\frac{62}{63}\right) \approx 0.0231$ bits.

**But what happens if my first question is:**

- $<$ **32 is wrong:** New information is $-\log_2\left(\frac{32}{64}\right) = 1$ bit!

## Entropy $E(I(X))$

We quantify the messiness of an input using the variable **Entropy.**

$$H(X) = -\sum_{i=1}^{n} p(x_i)\log_2 p(x_i)$$

Looking familiar? Entropy is the **expected value** of information across all possible outcomes.

Entropy measures the **purity** or uncertainty in a dataset. Lower entropy means less uncertainty – data is pure.

## Decision Tree

Decision Tree is a **Simple**, **Cheap To Run** Supervised Learning Algorithm that can perform tasks like classification, regression, and decision. We can use this to automatically:

- Shortlist interviewees from a pool of applicants
- Predict house prices
- Diagnose lung cancer based on X-Ray scan
- Classify Spam Email

While it cannot perform complex task like ChatGPT, it is useful in domains where **explainability** is critical, or **computational power** is limited.

**A typical decision looks like this:**

If Age $>$ 50 and Cholesterol $>$ 200 and $\ldots > \ldots \rightarrow$ High Risk

## Logic of Algorithm

Our input is a dataset: A table that has multiple features of a variable/example.

| Age | Income | Credit Score | Approved |
|---|---|---|---|
| 25 | Low | 600 | No |
| 45 | High | 750 | Yes |
| 30 | Medium | 680 | Yes |
| 22 | Low | 550 | No |

Table: Example dataset for loan approval

The last column "Approved" is what we are trying to **predict**. This is called the **target**, or the **output** of our algorithm.

Therefore a decision tree is a series of yes/no questions that split the dataset (Figure 1). We look through all features and find the one that **splits the data best.**

We use the **Entropy** quantity to determine which feature splits the data best. The **best feature split** (question) comes **first** in the algorithm.

The process of finding the question that decreases entropy the most is called **Information Gain**, the core of our Machine Learning algorithm.

## References


- Brownlee, Jason. "Information Gain and Mutual Information for Machine Learning." Machine Learning Mastery, 15 Oct. 2019, machinelearningmastery.com/information-gain-and-mutual-information/.
- Gray, Robert M. *Entropy and Information Theory*, First Edition, Corrected. 26 June 2023.
- Hankare, Omkar. "Decision Trees." *Medium*, 29 Jan. 2023, ompramod.medium.com/decision-trees-6a3c05e9cb82. Accessed 30 June 2025.

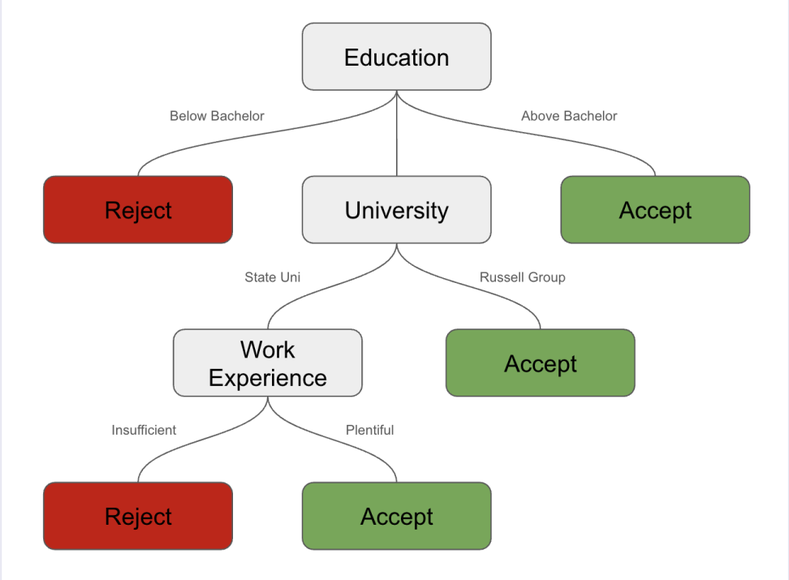

## Information Gain $H(D)$

Information gain measures the reduction in entropy after splitting an attribute. It is the difference between the entropy of the **original dataset** and the **weighted average entropy** after the dataset is split on attribute A. This quantifies how much **a feature** contributes to reducing uncertainty of the **entire dataset**.

$$\text{Information Gain}(D, A) = H(D) - \sum_{v \in \text{Values}(A)} \frac{|D_v|}{|D|} \cdot H(D_v)$$

- $D$: the full dataset at the current node
- $A$: the attribute used to split the data
- Values($A$): the set of all possible values of attribute $A$
- $\frac{|D_v|}{|D|}$: the proportion (weight) of examples in subset $D_v$
- $H(D)$: the entropy of the dataset before splitting
- $H(D_v)$: the entropy of subset $D_v$

In decision tree algorithms, information gain is used to choose the optimal feature that splits each node.

We aim to **maximize information gain** at each split to **reduce uncertainty** as much as possible. This **increases the purity** of the resulting subsets, allowing the decision tree to classify inputs more accurately and efficiently.

## Figure 1 - Example of Tree Split Diagram for Shortlisting Interviewees